# LEO HERNANDEZ

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## SUMMARY

Results-driven **Senior Java Full Stack Engineer** with **10+ years** of experience designing and developing high-performance cloud-based, microservices-driven applications. Deep expertise in **Java frameworks** and **full-stack development**, with strong frontend capabilities in **React**, **Angular**, and modern JavaScript frameworks. Specialized in bridging backend Java services with responsive, user-focused frontend interfaces for seamless application experiences. Demonstrated expertise in microservices architecture, RESTful APIs, and cloud platforms such as AWS, Azure and GCP. Skilled in utilizing modern development tools and methodologies like Agile, DevOps, and CI/CD to deliver robust software solutions. Proven track record delivering mission-critical applications across **Healthcare**, **Financial services**, and logistics industries with exceptional technical depth in concurrent programming, distributed systems, and performance optimization.

## EXPERIENCE

**Senior Full Stack Engineer / Software Architect** **01/2022 - 03/2025**

**Datavant** | Atlanta, GA | Hybrid

- Led the development of comprehensive healthcare interoperability platform using **Java 17** and Spring Boot 3.0, improving data exchange capabilities by 40% while ensuring HIPAA compliance and reducing integration time by 35%.
- Designed and implemented FHIR-compliant API gateway using Spring Cloud Gateway with custom GlobalFilters and RoutePredicates, enabling secure and efficient healthcare data exchange across disparate systems.
- Engineered robust event-driven architecture using **Apache Kafka** with Avro schemas and Schema Registry, implementing exactly-once semantics and transactional outbox pattern that reduced data inconsistencies by 65%.
- Implemented Kafka Streams topology for real-time healthcare analytics, processing 25,000+ events per second with sub-millisecond latency using custom processor implementations and windowed aggregations.
- Designed and optimized **Oracle** database solutions including complex stored procedures, functions, and triggers, implementing performance tuning techniques such as query optimization, indexing strategies, and partition schemes that improved database response time by 35%.
- Developed custom Kafka Connect connectors for healthcare systems integration, enabling seamless data flow between legacy systems and modern cloud platforms while maintaining compliance with industry regulations.
- Created specialized partitioning strategies for Kafka topics based on patient data, optimizing throughput and ensuring data locality for improved query performance across healthcare domains.
- Migrated legacy healthcare systems to cloud-native architecture, reducing infrastructure costs by 35% while improving system reliability through containerization and **Kubernetes** orchestration.
- Designed and implemented multi-region deployment architecture with active-active configuration, achieving 99.99% system availability and comprehensive disaster recovery capabilities.
- Implemented CQRS pattern with dedicated command and query databases (PostgreSQL for commands, Elasticsearch for queries), improving query performance by 42% while maintaining strict data consistency.
- Developed custom Spring Data repositories with specialized query methods and projections, reducing database load and optimizing application performance for healthcare data retrieval patterns.
- Engineered distributed task scheduling system using Quartz with JDBC JobStore, implementing clustering mechanisms that enabled zero-downtime maintenance and 99.9% job execution reliability.
- Created custom job execution listeners and triggers for healthcare data processing workflows, enabling sophisticated batch processing with automatic retry and compensation mechanisms.
- Developed comprehensive security framework using Spring Security with OAuth2/OpenID Connect, implementing custom authentication providers and role-based access control that satisfied stringent healthcare security requirements.

- Implemented fine-grained authorization using custom permission evaluators and SpEL expressions, ensuring precise access control for sensitive healthcare data across microservices.
- Created advanced monitoring solution using Micrometer and Prometheus, reducing mean time to detection (MTTD) of production issues by 65% through proactive alerting and comprehensive system visibility.
- Developed custom Grafana dashboards with automated anomaly detection, enabling real-time monitoring of system health and performance across distributed healthcare applications.
- Engineered resilience patterns using Resilience4j, implementing circuit breakers, bulkheads, and retry mechanisms that improved system fault tolerance by 78% during partial outages.
- Created comprehensive distributed tracing solution using Spring Cloud Sleuth and Zipkin, enabling precise identification of performance bottlenecks across microservice interactions.
- Implemented custom SpanReporters and sampling strategies, optimizing tracing overhead while maintaining visibility into critical system components and data flows.
- Implemented GitOps workflow using ArgoCD with custom sync policies and rollback strategies, reducing deployment failures by 45% and enabling consistent environment management.
- Integrated HashiCorp Vault for secrets management and certificate rotation, implementing secure credential handling across Kubernetes clusters for healthcare applications.
- Developed healthcare data anonymization pipeline processing **1TB+ daily data** while ensuring regulatory compliance and maintaining analytical value for research applications.
- Implemented custom tokenization and data masking techniques for PHI (Protected Health Information), enabling secure data sharing while preserving statistical significance for research.
- Optimized system performance through targeted improvements including database query tuning, JVM memory management, and custom caching strategies, reducing response times by 45% under peak load.
- Implemented database sharding and partitioning strategies for PostgreSQL, enabling horizontal scaling of patient data while maintaining transactional integrity across database nodes.
- Created advanced caching solution using **Redis** with customized eviction policies and cache invalidation strategies, reducing database load by 50% for frequently accessed healthcare records.
- Implemented distributed locking mechanisms using Redisson, ensuring data consistency for concurrent operations across microservices without performance degradation.
- Developed custom data synchronization framework for multi-region deployments, enabling active-active configuration with conflict resolution strategies for healthcare data.
- Implemented reactive programming models using Project Reactor, creating non-blocking data processing pipelines that improved throughput by 35% for high-volume healthcare data streams.
- Engineered custom Flux and Mono operators for healthcare-specific data transformation, optimizing memory utilization while maintaining backpressure handling for large datasets.
- Created automated FHIR resource validation framework, ensuring data quality and compliance with healthcare interoperability standards across all API endpoints.
- Implemented comprehensive **HIPAA compliance** framework with audit logging, data encryption, and access control mechanisms, ensuring regulatory compliance across all system components.
- Developed specialized logging framework using Log4j2 with contextual logging (MDC), implementing structured logging patterns optimized for security audit and compliance reporting.
- Created advanced full-text search capabilities using Elasticsearch with custom analyzers and tokenizers optimized for medical terminology and healthcare taxonomies.
- Designed and implemented schema evolution strategies for Avro and JSON data models, enabling backward and forward compatibility across service versions without downtime.
- Collaborated with frontend team to integrate **React** components with Java backend services, implementing Redux Toolkit state management patterns and type-safe API integrations for a cohesive user experience.
- Developed reusable API client libraries in TypeScript, reducing frontend integration time by 40% while ensuring consistent error handling and request formatting.

- Created specialized API response models with HATEOAS principles, enabling discoverable and self-documenting APIs that simplified frontend integration and reduced coupling.
- Implemented server-sent events (SSE) and WebSocket connections for real-time data updates, enabling live monitoring dashboards for critical healthcare metrics.
- Designed and built responsive user interfaces using **React** and **TypeScript**, creating reusable component libraries that reduced frontend development time by 35% across multiple projects.
- Implemented client-side state management using **Redux** for complex healthcare dashboards.
- Developed advanced frontend error handling and recovery strategies, improving user experience and reducing support tickets by 30% for critical healthcare applications.
- Mentored team of 6 developers, conducting architecture design sessions and technical knowledge sharing, resulting in 30% reduction in production incidents and improved code quality metrics.
- Led migration from monolithic Jenkins pipelines to GitOps workflow with ArgoCD, improving deployment consistency and reducing release cycle time by 65%.
- Implemented automated architecture compliance verification using custom static analysis rules, enforcing design principles and preventing architectural drift across teams.

**Environment**: Java 8-17, Spring Boot, Spring Cloud, Spring Security, Spring Batch, Hibernate/JPA, Apache Kafka, Elasticsearch, Redis, PostgreSQL, MongoDB, React, Redux, TypeScript, Docker, Kubernetes, AWS, Azure, Microservices, RESTful APIs, GraphQL, OAuth2, JWT, CI/CD, Git, Maven/Gradle, Jenkins, ArgoCD, Prometheus, Grafana, Zipkin, ELK Stack

**Lead Java Developer** **03/2019 – 12/2021**
**Tyler Technologies** | Plano, TX | Remote

- Engineered high-throughput transaction processing system using **Java 11** and Spring Framework, processing 15,000+ transactions per second with sub-millisecond latency and 99.999% reliability.
- Developed comprehensive batch processing framework using Spring Batch with partitioned job execution, reducing processing time for financial data by 65% while ensuring transactional integrity.
- Created custom JobRepository implementation with MongoDB for metadata storage, implementing optimistic locking for concurrent job execution and custom JobExplorer for operational monitoring.
- Implemented data access layer using Spring Data JPA with custom repositories, query methods, specifications, and projections, utilizing composite repositories with multiple persistence technologies.
- Developed distributed search solution using Elasticsearch with custom analyzers, tokenizers, and query builders, implementing parent-child relationships, nested documents, and custom routing strategies for financial data retrieval.
- Created comprehensive logging framework using Log4j2 with asynchronous logging using Disruptor, implementing custom appenders, layout patterns, and filters with contextual logging using MDC.
- Implemented centralized log aggregation using ELK stack with custom Logstash filters and Elasticsearch index templates, creating Kibana dashboards with custom visualizations for operational monitoring.
- data capture (CDC) solution using Debezium with custom SMTs (Single Message Transforms) for data transformation and filtering, implementing outbox pattern for reliable event publishing.
- Created distributed caching solution using Hazelcast IMDG with near-cache configurations, implementing entry processors for in-place data modifications, partition-aware operations, and split-brain protection.
- Implemented custom serialization strategies for optimized network transfer, utilizing compact serialization format and delta updates for bandwidth optimization.
- Developed comprehensive testing strategy using JUnit 5, Mockito, and TestContainers, implementing parameterized tests, custom extensions, and dynamic tests with test data generators.
- Led team of 12 developers across 3 scrum teams, conducting architecture reviews, technical design sessions, and mentoring activities, resulting in 40% reduction in technical debt and improved system reliability.

**Environment**: Java 11, Spring Framework, Spring Batch, Spring Data JPA, MongoDB, Elasticsearch, Apache Camel, Debezium, Hazelcast, JUnit 5, Mockito, TestContainers, ELK Stack, Kibana, Logstash, Agile/Scrum

**Java Full Stack Developer** **06/2017 to 02/2019**
**SolarWinds |** Austin, TX | Onsite

- Engineered modular monolith application using Java 8 and Spring Framework with aspect-oriented programming for cross-cutting concerns, implementing custom BeanFactoryPostProcessor and BeanPostProcessor for runtime bean modifications.
- Developed comprehensive data access layer using JPA/Hibernate with custom UserType implementations, interceptors, and entity listeners, implementing composite key mappings, inheritance strategies, and optimistic locking mechanisms.
- Collaborated on implementing WebSocket connections for real-time data updates between Java backend and JavaScript frontend components.
- Created custom dialect extensions for PostgreSQL with native query optimizations, implementing custom functions, operators, and type mappings for improved performance.
- Implemented RESTful API using Spring MVC with custom HandlerMethodArgumentResolver, ReturnValueHandler, and message converters, creating hypermedia-driven API with HATEOAS principles.
- Developed custom BPMN extensions with domain-specific activities and gateways, implementing process migration strategies for in-flight process instances.
- Created LDAP integration with custom user details service and authority mapping, implementing single sign-on with SAML 2.0 and custom assertion validation.

**Environment**: Java 8, Spring Framework, Spring MVC, JPA/Hibernate, PostgreSQL, EhCache, JGroups, Spring Security, LDAP, ELK Stack, JMS, ActiveMQ, Atomikos XA, JUnit, Mockito, Apache POI, PDFBox, BPMN

**Java Full Stack Developer** **8/2014 to 07/2017**
**Datadog |** New York, NY | Onsite

## EDUCATION

**Bachelor of Science**: **Computer Science** **05/2014**
Georgia State University Atlanta, GA, United States

## SKILLS

- **Programming Languages**: Java (8-17), Kotlin, Groovy, JavaScript/TypeScript, SQL (T-SQL, PL/SQL), Python, Bash/Shell, XML, JSON, GraphQL
- **Java & Backend Frameworks**: Spring Framework, Spring Boot (2.x, 3.x), Spring Cloud, Spring Security, Spring Data JPA, Spring Batch, Spring Integration, Spring WebFlux, Hibernate/JPA, Project Reactor, RxJava, Axon Framework, Apache Camel, Quartz
- **Messaging & Integration**: Apache Kafka, Kafka Streams, Kafka Connect, RabbitMQ, ActiveMQ, JMS, Apache Camel, WebSockets, gRPC
- **Databases**: Oracle (11g, 12c, 19c), PostgreSQL, MongoDB, MySQL/MariaDB, Redis, Elasticsearch, Cassandra, DynamoDB, JDBC, Connection Pooling
- **Cloud & Infrastructure**: AWS (EC2, S3, Lambda, DynamoDB, RDS), Azure (AKS, App Service, Cosmos DB), GCP, Docker, Kubernetes, Helm, Terraform, Ansible
- **DevOps & CI/CD**: Jenkins, GitLab CI, GitHub Actions, CircleCI, ArgoCD, Maven, Gradle, SonarQube, Git, GitHub, GitLab
- **Monitoring & Observability**: ELK Stack, Prometheus, Grafana, Micrometer, Zipkin, Datadog, New Relic, Log4j2, Logback
- **Testing**: JUnit (4, 5), Mockito, TestContainers, REST-assured, Cucumber, JMeter, Gatling, TDD, BDD
- **Architecture**: Microservices, Event-Driven Architecture, Domain-Driven Design, CQRS, Hexagonal Architecture, RESTful APIs, GraphQL
- **Security**: OAuth2/OpenID Connect, JWT, SAML, LDAP, Spring Security, Encryption, Role-Based Access Control
- **Frontend**: React, Redux, Angular (2+), NgRx, Vue.js, TypeScript, HTML5, CSS3, SCSS, Webpack, Jest, React Testing Library, Bootstrap, Material-UI

# Leo Hernandez

- Norcross, GA, US

## Contact Information


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- 6785683304


## Work History

### Total Work Experience: 11 years

- **Senior Full Stack Engineer Datavant**
  Jan 01, 2022
- **Lead Java Developer Tyler Technologies**
  Mar 01, 2019
- **Java Full Stack Developer Solarwinds**
  Jun 01, 2017
- **Java Full Stack Developer Datadog**
  Aug 01, 2014

## Education

- **Bachelors** | Georgia State University

## Skills

- **api** - 11 years
- **java** - 11 years
- **ui** - 11 years

- **kotlin -** 10 years
- **spring framework -** 10 years
- **sql -** 10 years
- **architecture -** 9 years
- **aws -** 9 years
- **elasticsearch -** 9 years
- **framework -** 9 years
- **graphql -** 9 years
- **integration -** 9 years
- **qa -** 9 years
- **spring -** 9 years
- **strategy -** 9 years
- **transformation -** 9 years
- **jenkins -** 8 years
- **jpa -** 8 years
- **mockito -** 8 years
- **policies -** 8 years
- **apache kafka -** 7 years
- **documentation -** 7 years
- **leadership -** 7 years
- **monitoring -** 7 years
- **operations -** 7 years
- **routing -** 7 years
- **specification -** 7 years
- **batch processing -** 6 years
- **caching -** 6 years
- **configuration -** 6 years
- **dashboard -** 6 years
- **mongodb -** 6 years
- **reliability engineering -** 6 years
- **spring batch -** 6 years
- **websocket -** 5 years
- **restful -** 8 years
- **layout -** 7 years

## Work Preferences

- Desired Work Settings: Remote
- Likely to Switch: True
- Willing to Relocate: False
- Travel Preference: 0%
- Work Authorization:
    - US

- Work Documents:
    - US Citizen

Security Clearance: False
- Desired Hourly Rate: 65+ (USD)
- Desired Salary: 125,000+ (USD)
- Third Party: False
- Employment Type:
    - Full-time
    - Part-time

## Profile Sources

- Dice: https://www.dice.com/employer/talent/profile/0ee873d3db94c3eac459333bdf3a35bd